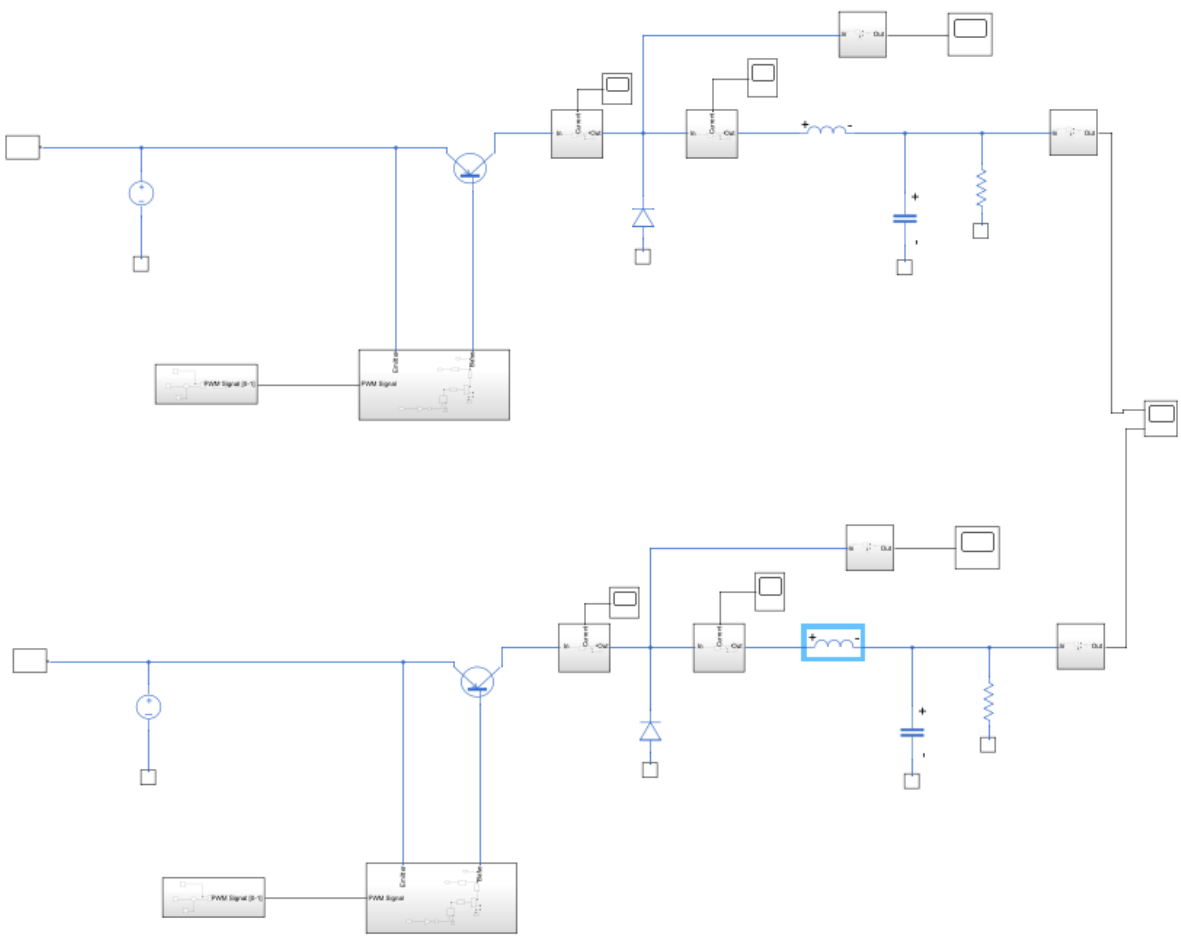
Power Electronics-Part4

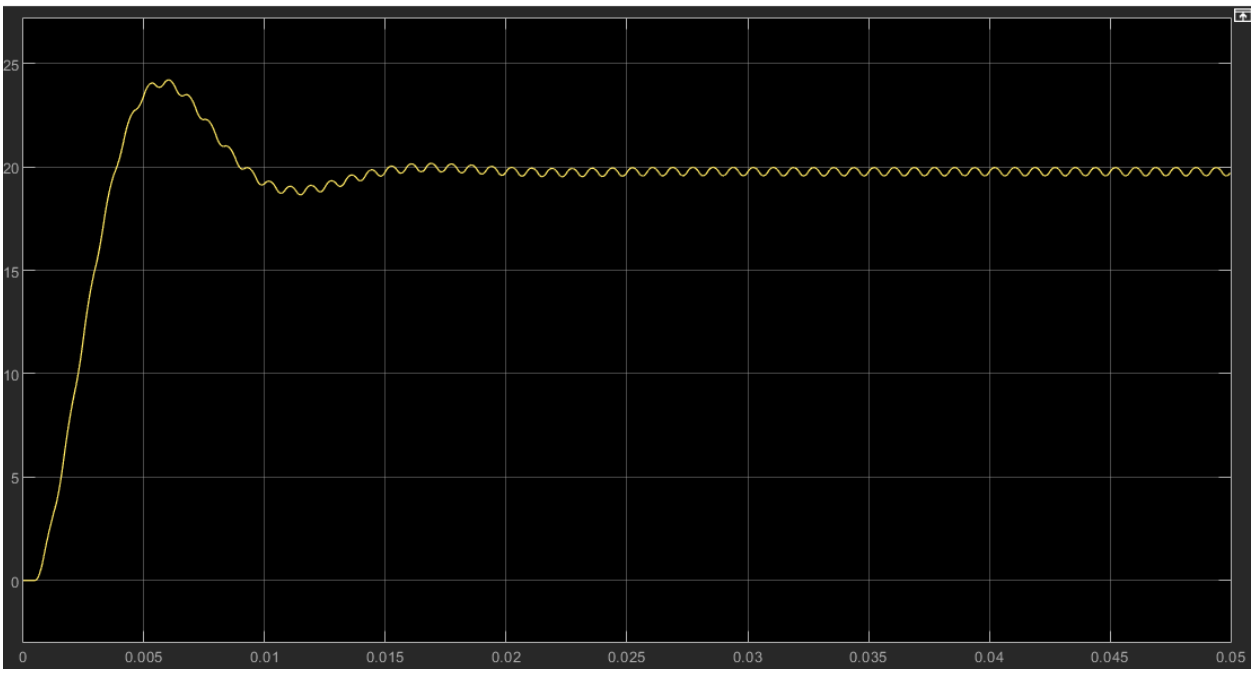
For this part I wanted to observe better the difference between the a) ideal case and b) the real case, so I doubled the schematic in Simulink (the top one for a), the bottom one for b)). (**the yellow signal is for a) and the blue one is for b)**)



a) Ideal simulation

In this case the 2 signals overlap perfectly because I haven't introduced the residual properties, nor the tolerances.

This is the output voltage in the ideal case

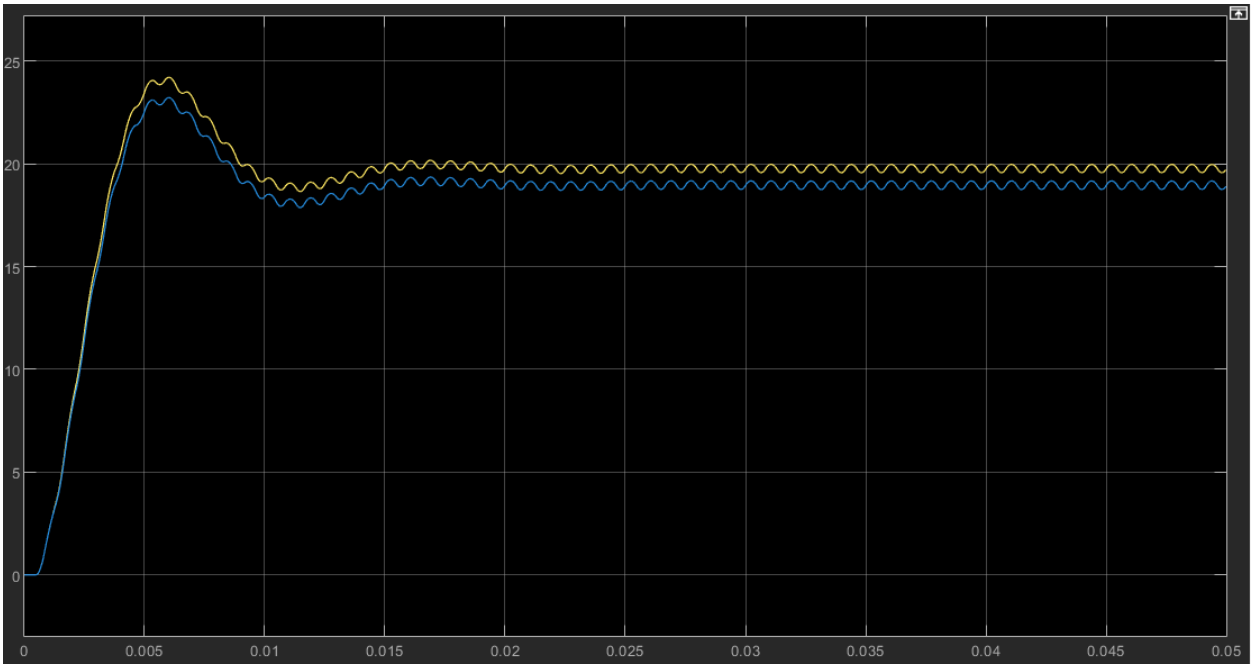


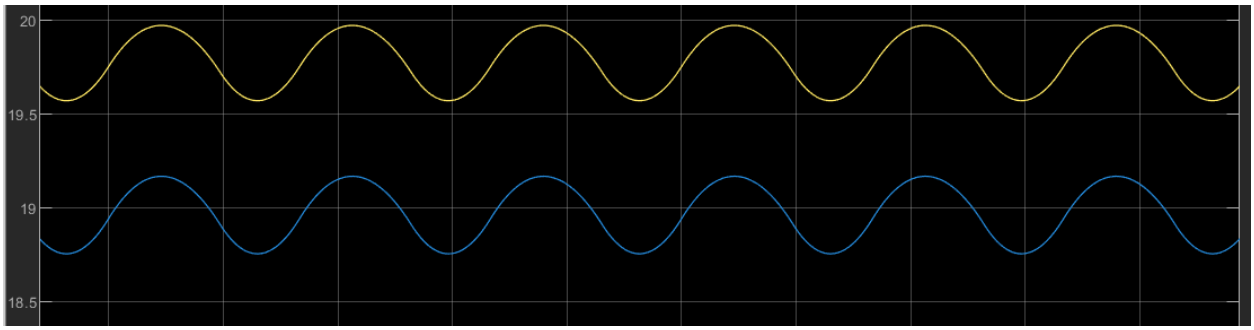
It has a steady state value of about 19.8 V with a $\pm$0.2V ripple and a 21% overshoot

b) With residual elements

Now I will start introducing the changes from the datasheet one by one and compare them with the ideal case

i) Diode – Forward voltage of 1.5 V (> 0.1 V from a))





The signal keeps its shape but is 0.8V smaller.

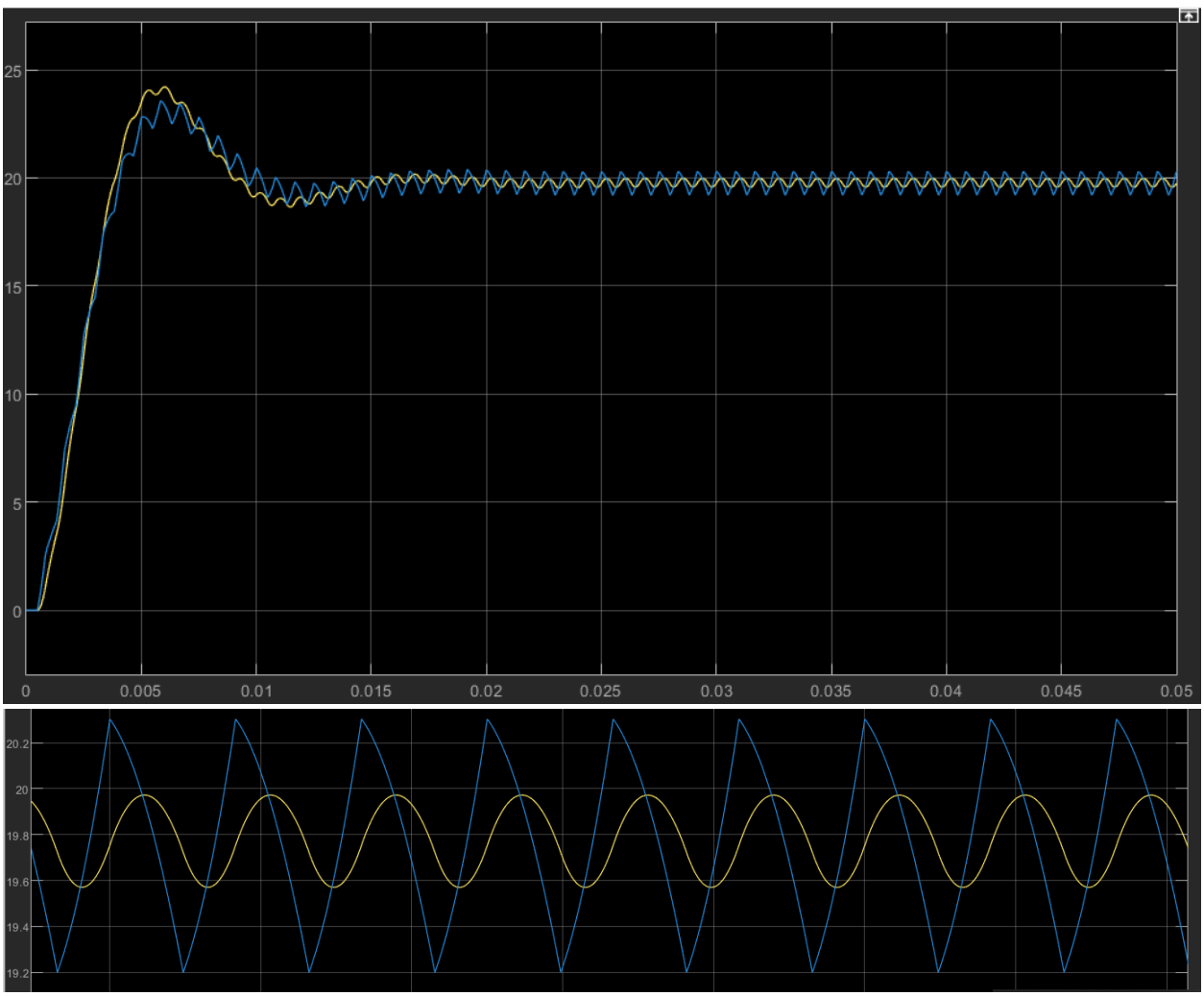
ii) Capacitor: Series resistance: ESR (I used the formula below)

$$ESR = \frac{tan(\tau)}{2 * \pi * f_{test} * C}$$

$tan(\tau) = 0.25\ (max\ value)$

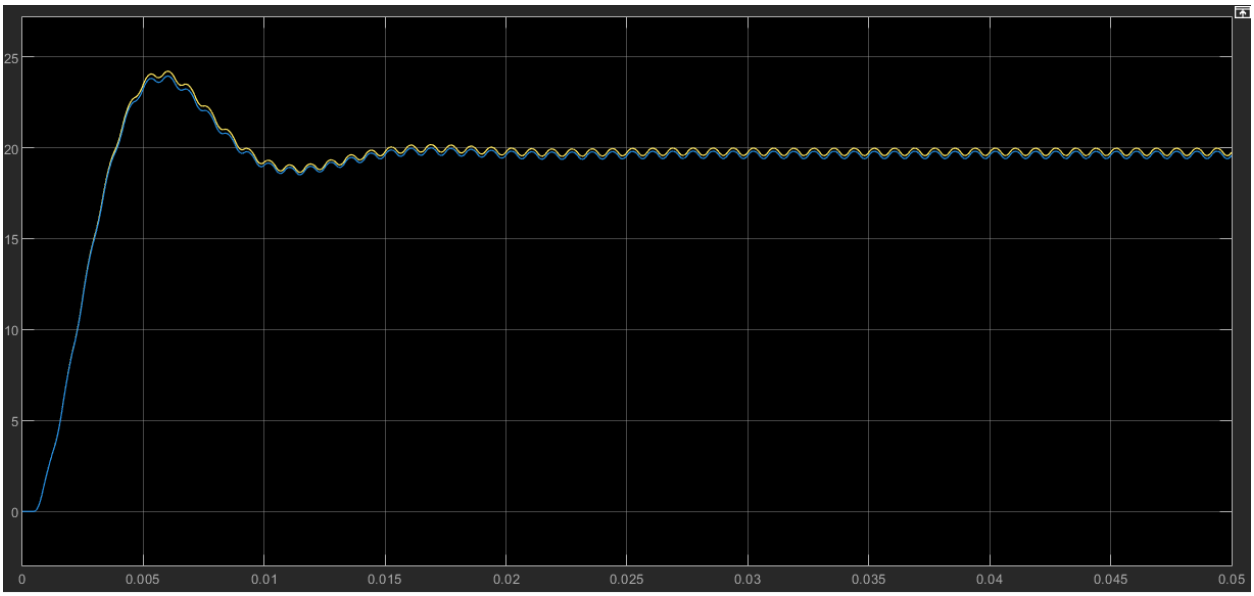
$f_{test} = 120\ Hz$ => $ESR = 0.1461\ \Omega$

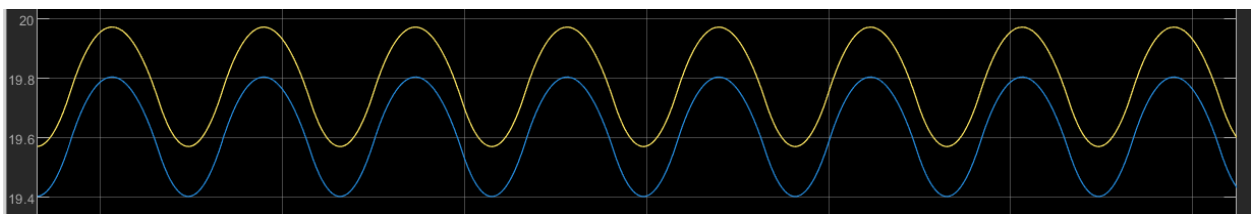
$C = 2.27\ mF$



We can clearly see the capacitor charging and discharging. The output is slightly shifted, has a ± 0.3 V variation from the original and a smaller ~ 16% overshoot.

iii) Inductor: Series resistance: DC resistance = 6.9 mΩ (from datasheet)

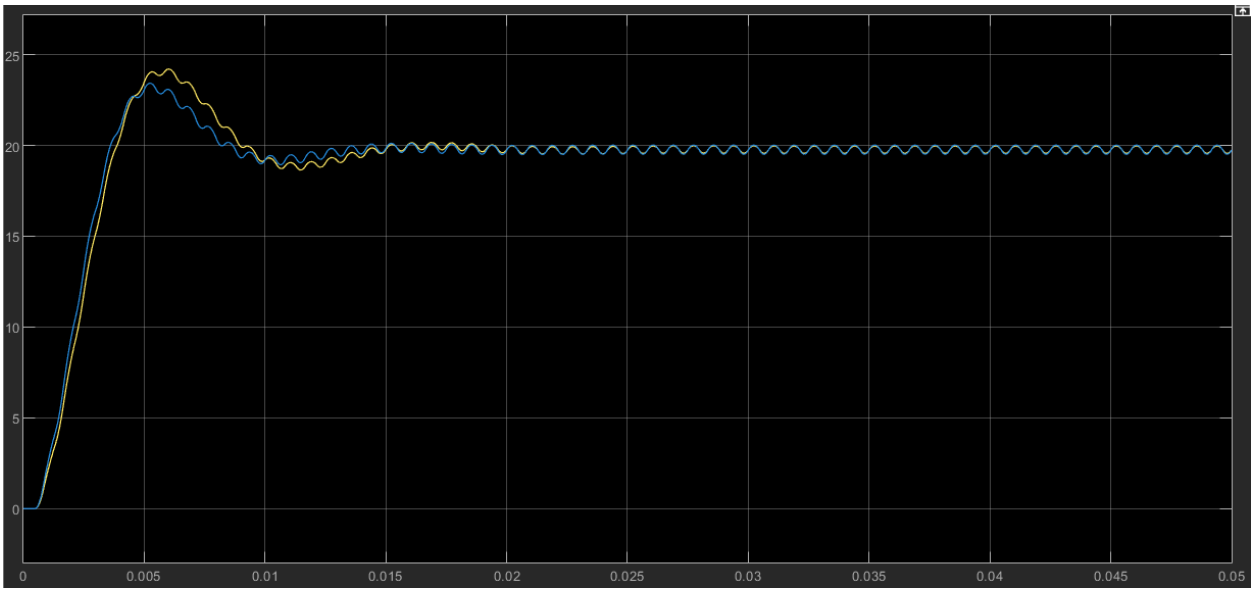




The shape and phase remain the same, only a 0.1 V drop in the overall output voltage

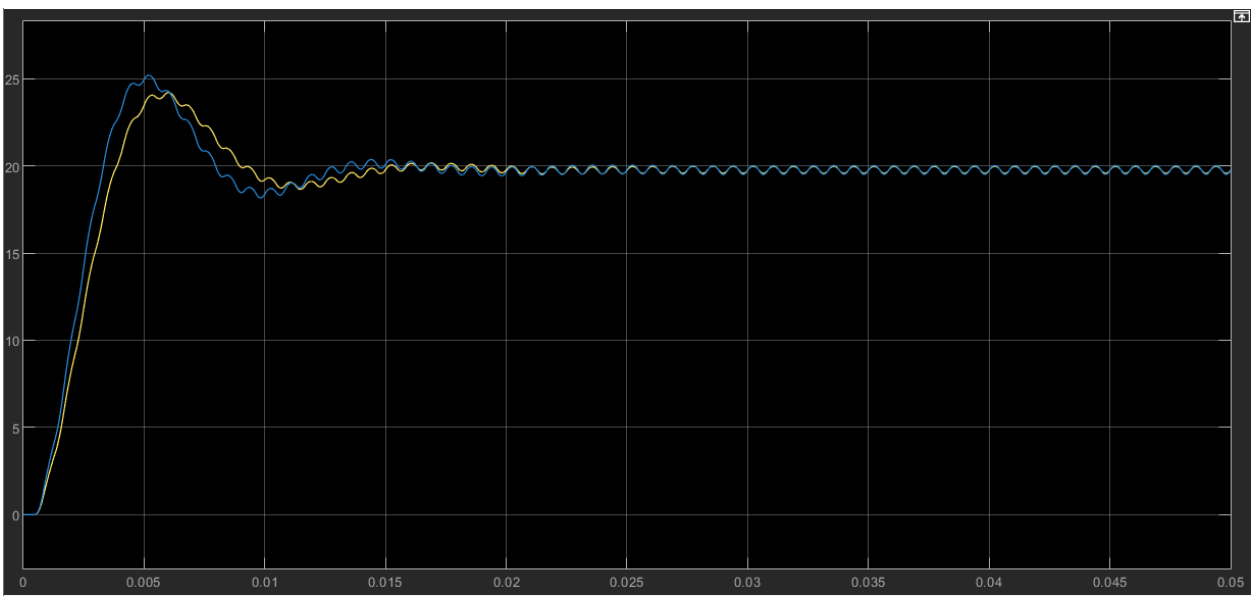
iv) Capacitor: Tolerance application = 20% (from datasheet)

I applied the minimum value, so I ended up with 80% of the initial capacitance



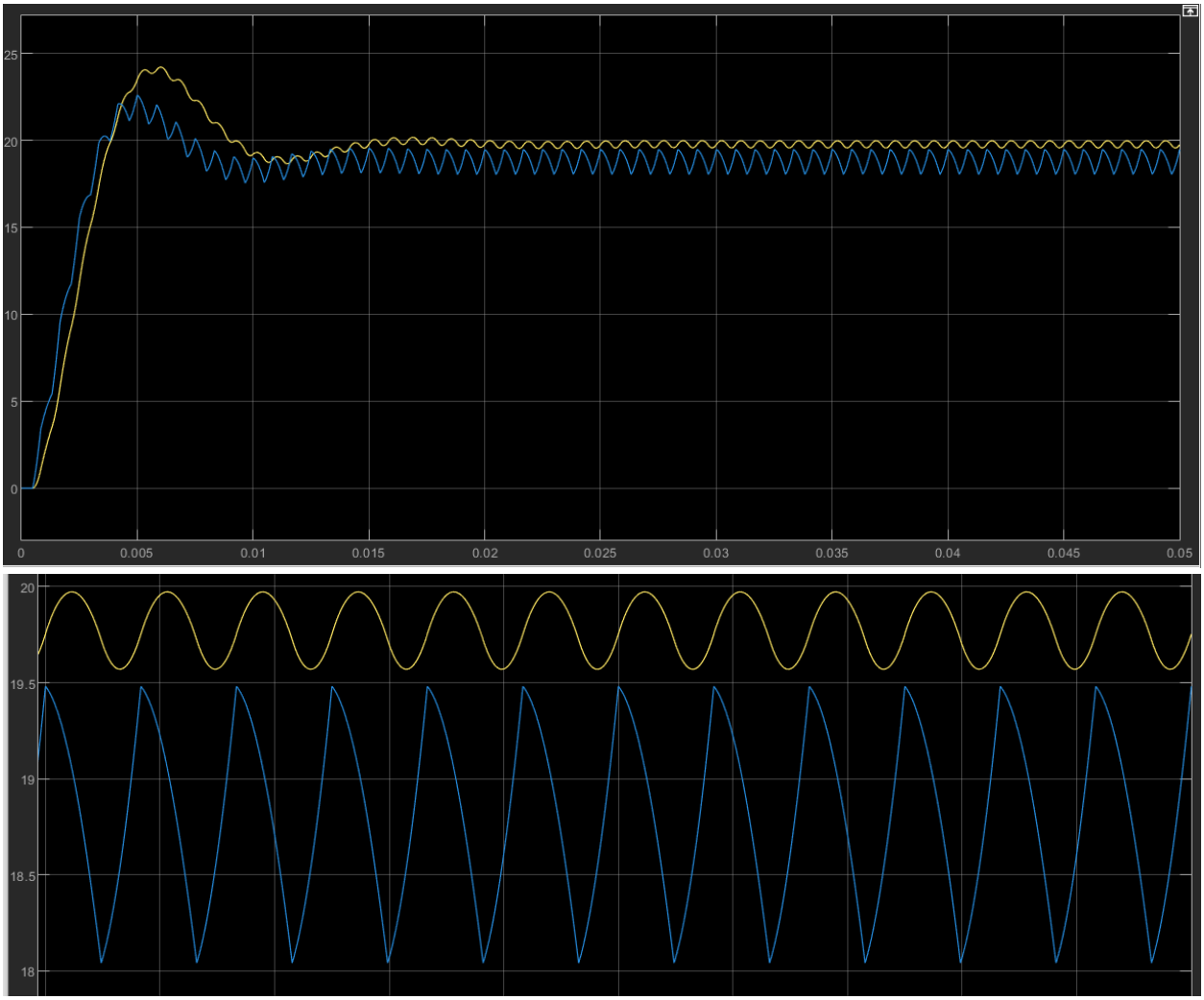
The output has a smaller overshoot of about 17% and ± 0.05 V in comparison to the ideal output

v) Inductor: Tolerance application - Apply minimum tolerance value



In this case the overshoot is bigger of about 26% and the same steady state difference as in the previous case.

Now I will implement all these residual elements at once



The obtained output has a smaller overshoot of 11% and has the shape induced by the capacitor series resistance . In steady state there is a 1 V drop and a slight phase shift.